



Team Ramses

Members

Romain Teyssier (PU astro)

Jonathan Palafoutas (PU PACM)

Robel Geda (PU astro)

James Sunseri (PU astro)

Yue Pan (PU astro)

Jun-Yong Lee (PU astro)

Members (continued)

William Grogger (PU astro)

Eric Moseley (Stanford)

Mentors

Burlen Loring (Nvidia)

Guillaume Barnier (Nvidia)



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RAMSES

Adaptive Mesh Refinement for radiative MHD with self-gravity and PIC for astrophysical fluid flows

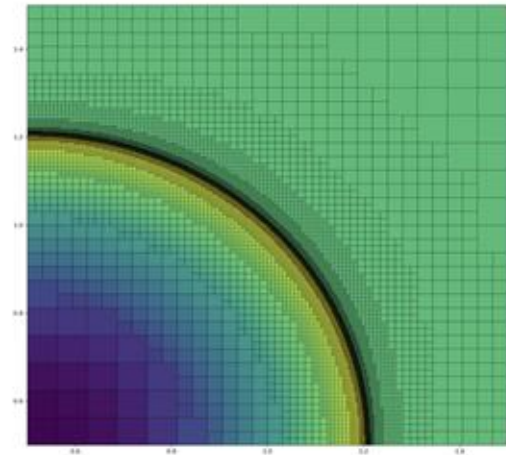
- PDE solvers using adaptive grids
- Cell-by-cell refinements
- MPI (only unigrid hydro for GPU)
- Nvidia Fortran (CUDA)



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<https://bitbucket.org/rteyssie/mini-ramses>





RAMSES on the GPU

CUDA Fortran with NVHPC Fortran compiler: `make COMPILER=NVHPC`

Single GPU execution for now. Data fully resident on the GPU.

Hash table on the GPU: `fnv64` hash function and `fast linear probing` for collisions.

Octs are sorted in memory first per refinement level, then per Hilbert key
Fast sorting on the GPU with `bucket sort` (level) and `LSD radix sort` (key).
Using prefix sum with `shfl_up/shfl_down` CUDA intrinsics

GPU kernels developed so far: hydro, AMR, self-gravity

GPU kernel in development: Particle-In-Cell



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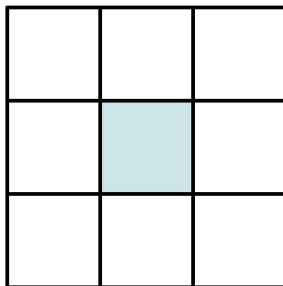
Hydro kernel geometry



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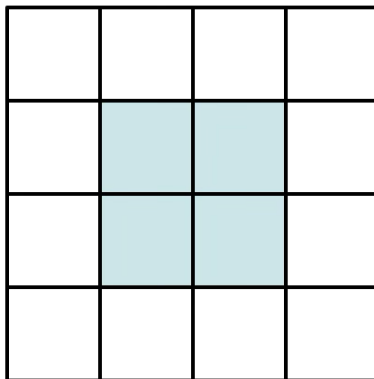


3x3x3



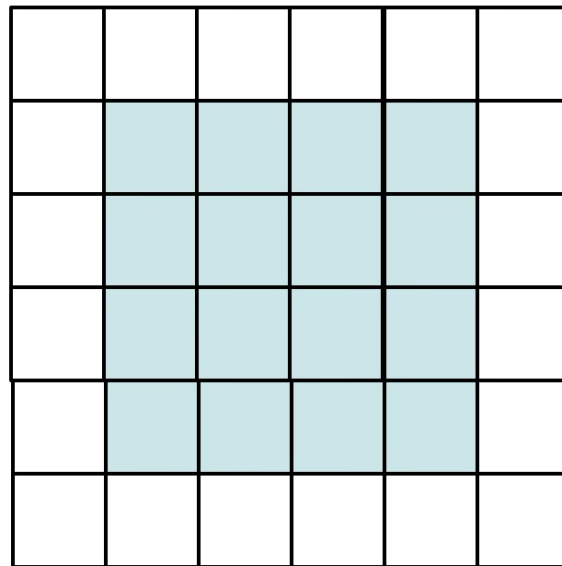
overhead=26
shared mem=12kB
block size=64

4x4x4



overhead=8
shared mem=41kB
block size=256

6x6x6



overhead=3
shared mem=210kB
block size=?



Roofline analysis of the hydro solver



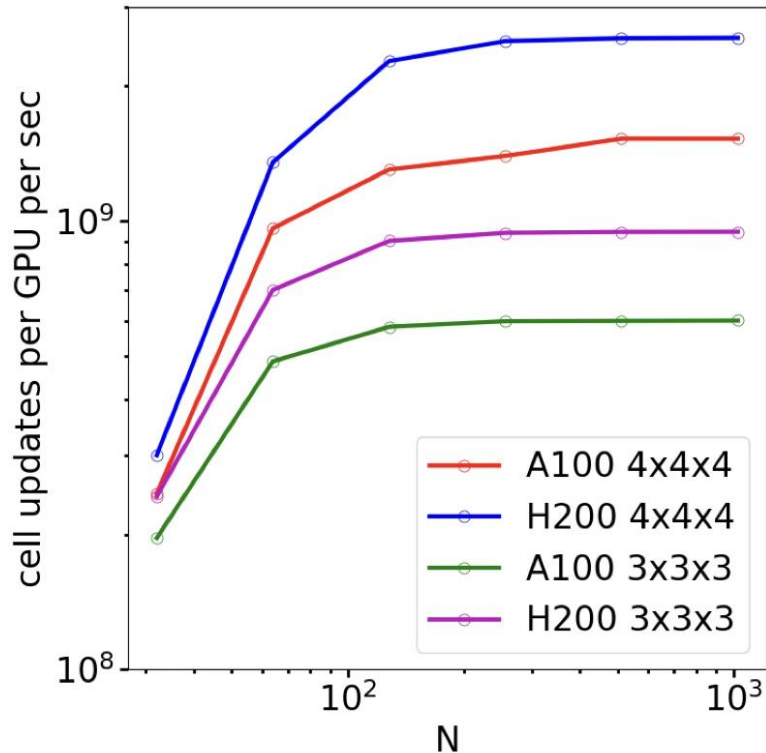
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Hydro solver with LLF
and cell unigrid with
single precision

Hydro kernel (Bob
Caddy) with 3x3x3 or
4x4x4 block geometry.

MPI 64 core
1e8 cell/node/sec





Our Goals for the Hackathon



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- Performance analysis AMR, MG and PIC
- GPU code optimization
- Finish PIC (Particle-In-Cell) solver
- Develop FMM (Fast Multipole Method) solver
- Add passive scalars to hydro solver
- Add MHD



Test cases we will use for the hackathon

- [sedov3d](#): point-explosion with and w/o AMR
- [coeur](#): collapse of molecular core with and w/o AMR.
- [dmo](#): N body cosmological simulation with AMR.



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